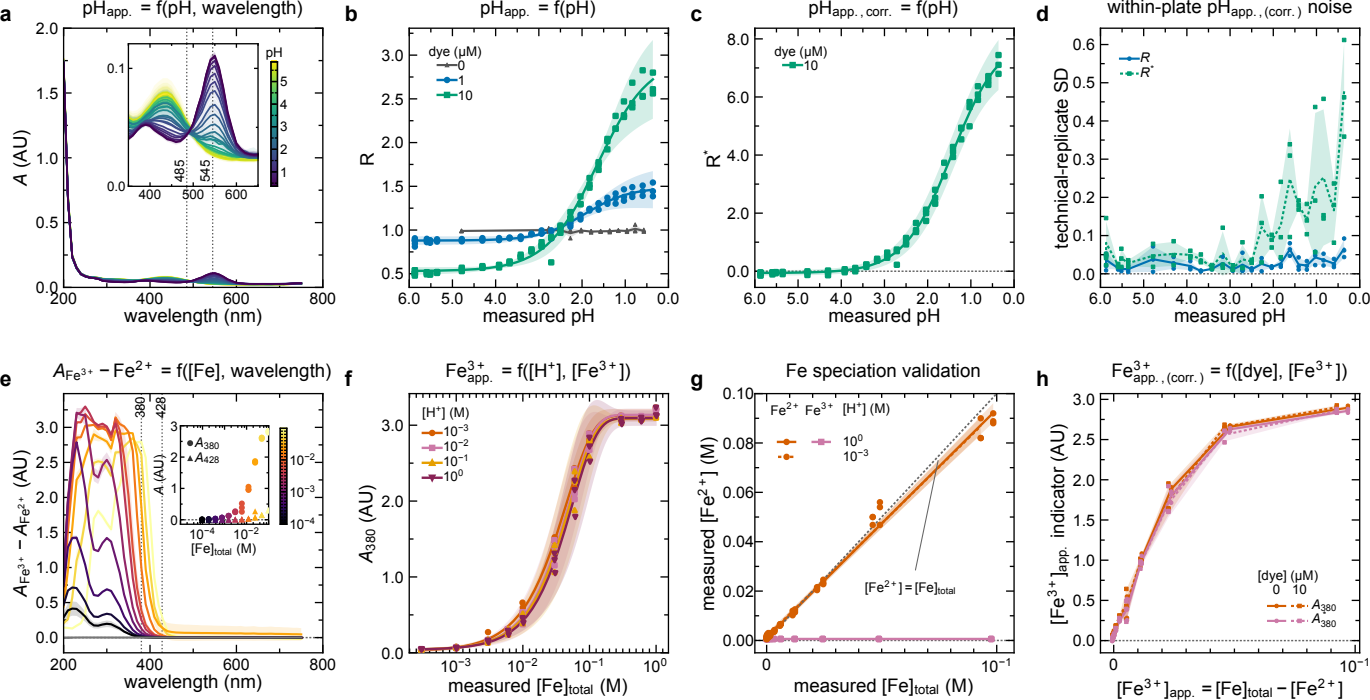
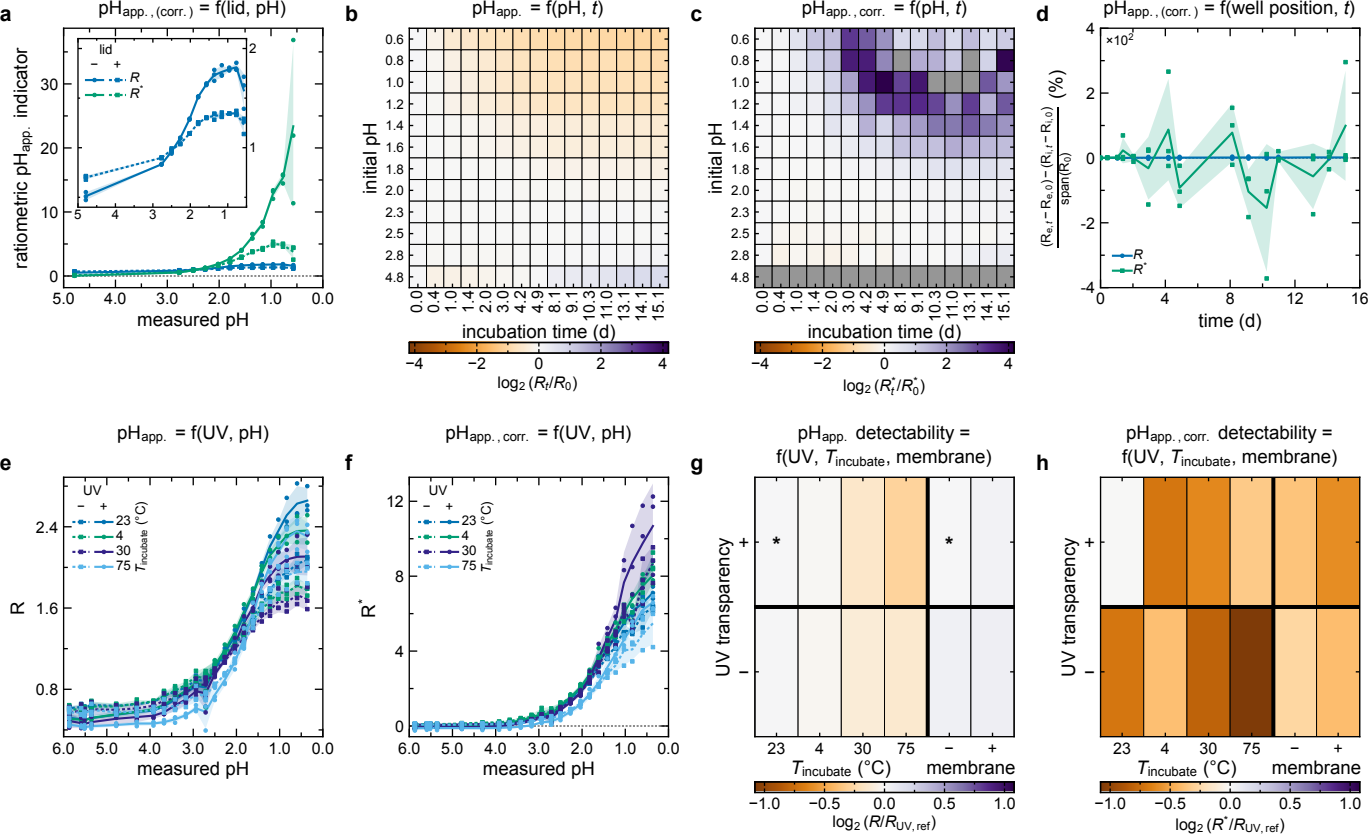
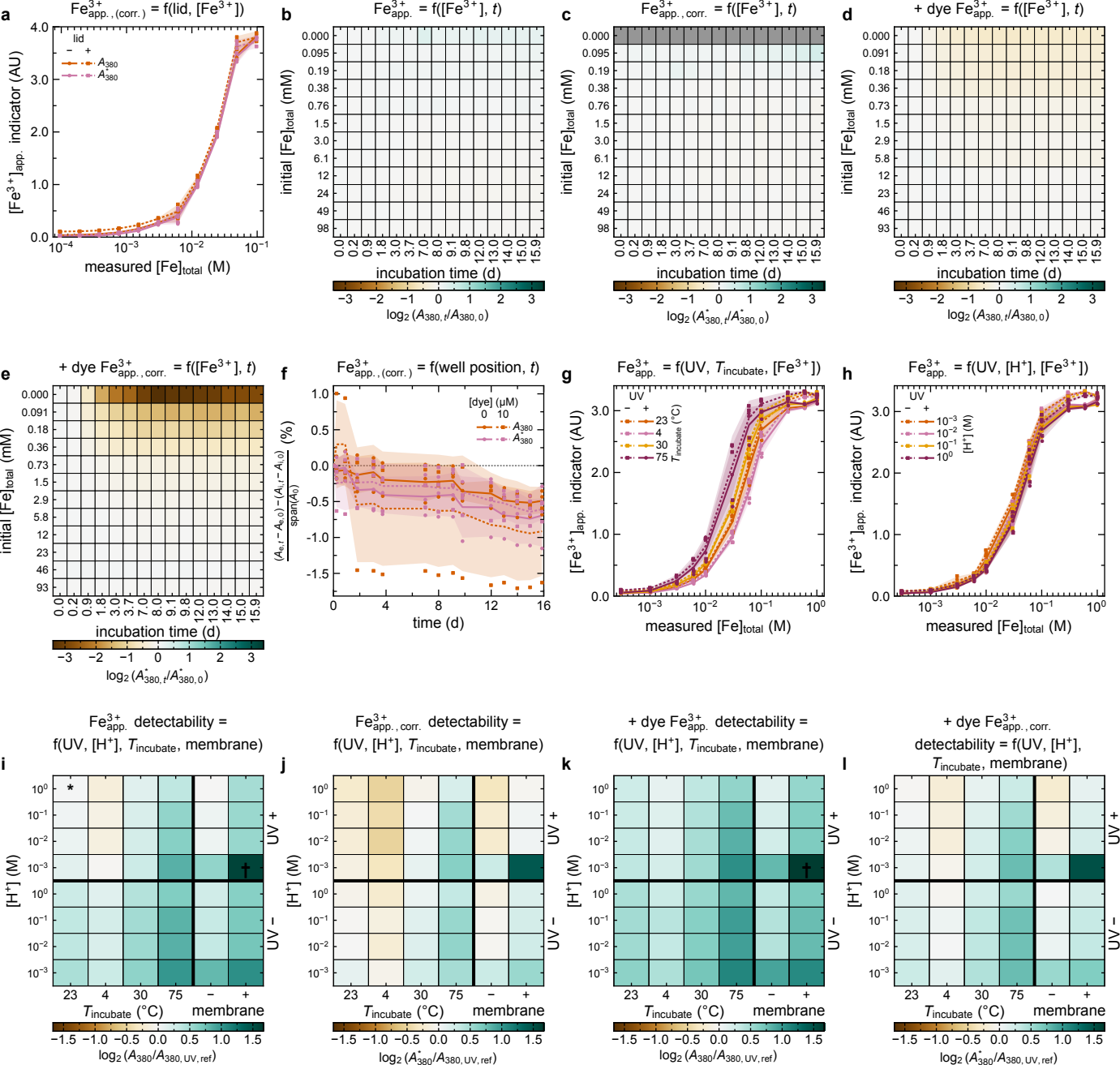
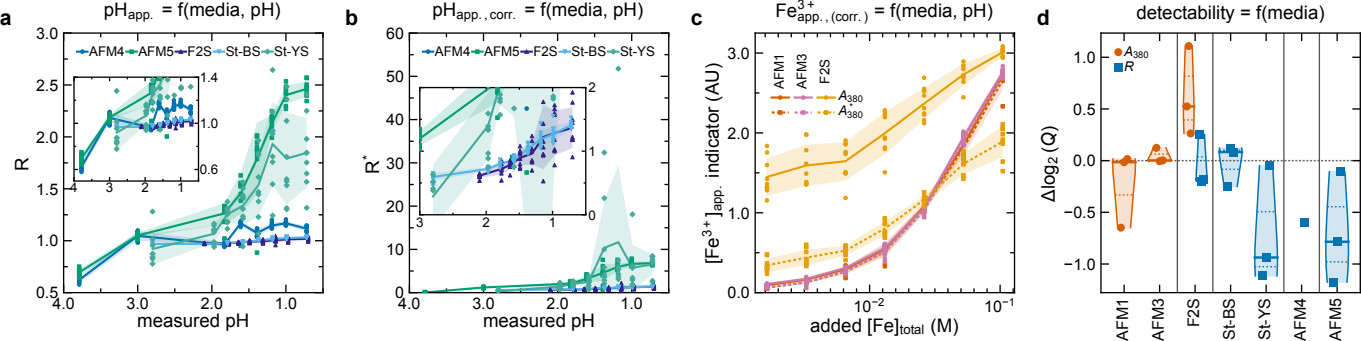
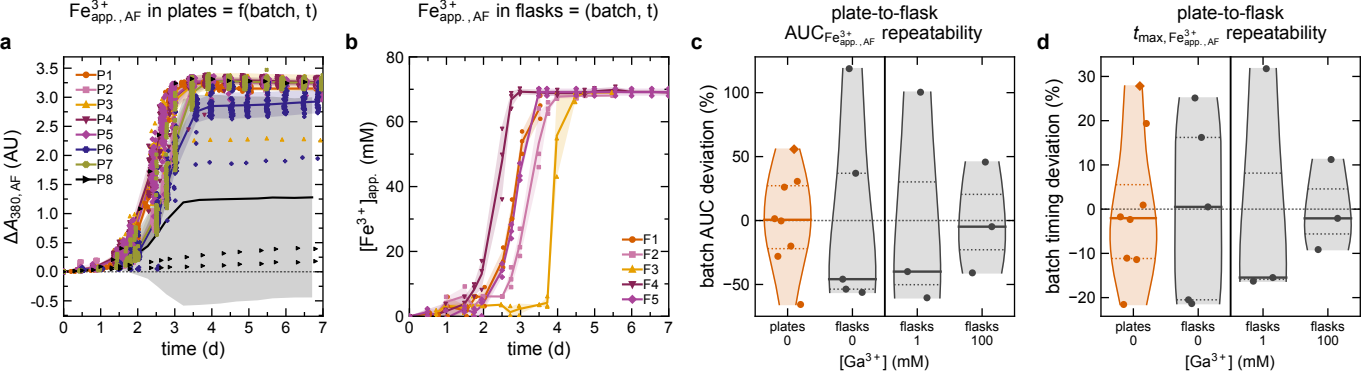
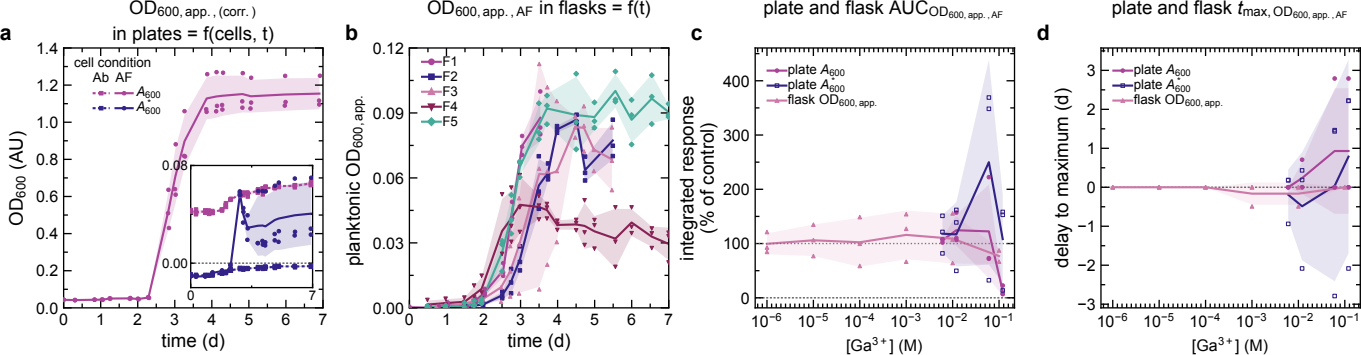

a
pH$_{app.}$ = f(pH, wavelength)
A (AU)
wavelength (nm)
pH
485
545
b
pH$_{app.}$ = f(pH)
dye (μM)
0
1
10
R
measured pH
c
pH$_{app.,corr.}$ = f(pH)
dye (μM)
10
R*
measured pH
d
within-plate pH$_{app.,(corr.)}$ noise
R
R*
technical-replicate SD
measured pH
e
$A_{Fe^{3+}} - Fe^{2+}$ = f([Fe], wavelength)
380
428
$A_{Fe^{3+}} - A_{Fe^{2+}}$ (AU)
wavelength (nm)
A_{380}
A_{428}
A (AU)
[Fe]$_{total}$ (M)
f
$Fe^{3+}_{app.}$ = f([H$^+$], [Fe^{3+}])
[H$^+$] (M)
10^{-3}
10^{-2}
10^{-1}
10^{0}
A_{380} (AU)
measured [Fe]$_{total}$ (M)
g
Fe speciation validation
Fe^{2+} Fe^{3+} [H$^+$] (M)
10^{0}
10^{-3}
measured [Fe^{2+}] (M)
[Fe^{2+}] = [Fe]$_{total}$
measured [Fe]$_{total}$ (M)
h
$Fe^{3+}_{app.,(corr.)}$ = f([dye], [Fe^{3+}])
[dye] (μM)
0 10
A_{380}
A_{380}
[Fe^{3+}]$_{app.}$ indicator (AU)
[Fe^{3+}]$_{app.}$ = [Fe]$_{total}$ − [Fe^{2+}]

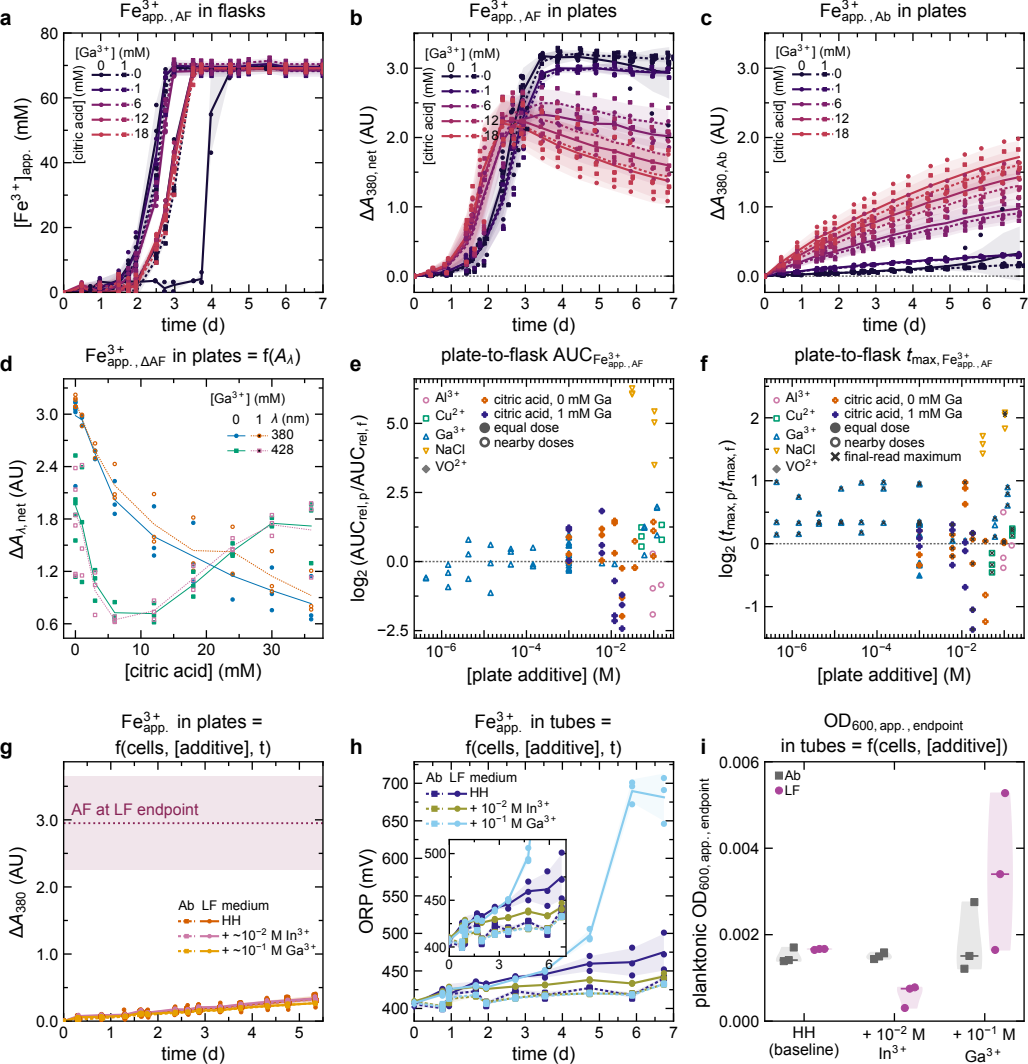

a
pH app., (corr.) = f(lid, pH)
ratiometric pH app. indicator
measured pH
lid
R
R*
b
pH app. = f(pH, t)
initial pH
incubation time (d)
log2 (Rt/R0)
c
pH app., corr. = f(pH, t)
initial pH
incubation time (d)
log2 (R*t/R*0)
d
pH app., (corr.) = f(well position, t)
time (d)
R
R*
e
pH app. = f(UV, pH)
UV
T incubate (°C)
R
measured pH
f
pH app., corr. = f(UV, pH)
UV
T incubate (°C)
R*
measured pH
g
pH app. detectability = f(UV, T incubate, membrane)
UV transparency
T incubate (°C)
membrane
log2 (R/RUV, ref)
h
pH app., corr. detectability = f(UV, T incubate, membrane)
UV transparency
T incubate (°C)
membrane
log2 (R*/RUV, ref)

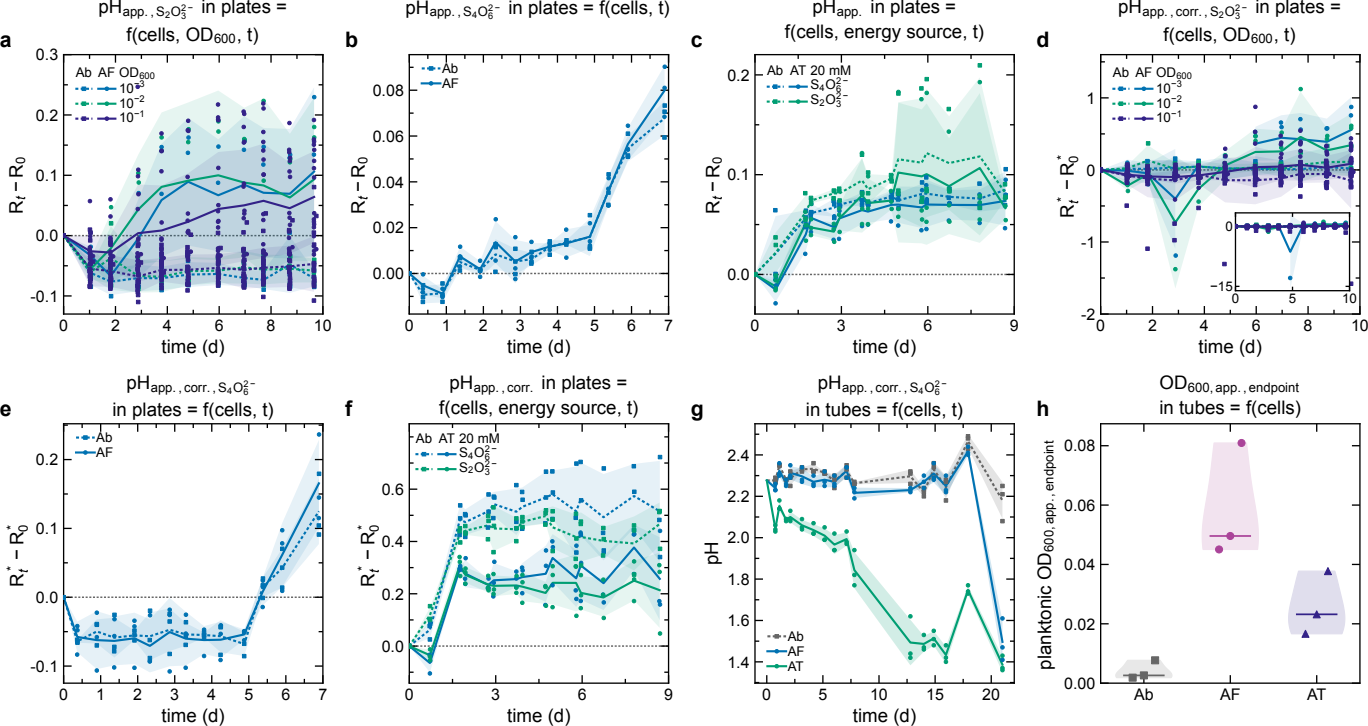

a
Fe3+app.,(corr.) = f(lid, [Fe3+])
lid
A380
A*380
[Fe3+]app. indicator (AU)
measured [Fe]total (M)
b
Fe3+app. = f([Fe3+], t)
initial [Fe]total (mM)
incubation time (d)
log2 (A380,t/A380,0)
c
Fe3+app.,corr. = f([Fe3+], t)
log2 (A*380,t/A*380,0)
d
+ dye Fe3+app. = f([Fe3+], t)
e
+ dye Fe3+app.,corr. = f([Fe3+], t)
f
Fe3+app.,(corr.) = f(well position, t)
[dye] (μM)
time (d)
g
Fe3+app. = f(UV, Tincubate, [Fe3+])
Tincubate (°C)
h
Fe3+app. = f(UV, [H+], [Fe3+])
[H+] (M)
i
Fe3+app. detectability = f(UV, [H+], Tincubate, membrane)
Tincubate (°C)
membrane
log2 (A380/A380,UV,ref)
j
Fe3+app.,corr. detectability = f(UV, [H+], Tincubate, membrane)
log2 (A*380/A*380,UV,ref)
k
+ dye Fe3+app. detectability = f(UV, [H+], Tincubate, membrane)
l
+ dye Fe3+app.,corr. detectability = f(UV, [H+], Tincubate, membrane)


a
pH_app. = f(media, pH)
AFM4
AFM5
F2S
St-BS
St-YS
R
measured pH
b
pH_app., corr. = f(media, pH)
R*
measured pH
c
Fe^3+_app., (corr.) = f(media, pH)
AFM1
AFM3
F2S
A380
A380
[Fe3+]app. indicator (AU)
added [Fe]total (M)
d
detectability = f(media)
A380
R
Δlog2 (Q)
AFM1
AFM3
F2S
St-BS
St-YS
AFM4
AFM5


a
$Fe^{3+}_{app.,AF}$ in plates = f(batch, t)
$\Delta A_{380,AF}$ (AU)
time (d)
P1
P2
P3
P4
P5
P6
P7
P8
b
$Fe^{3+}_{app.,AF}$ in flasks = (batch, t)
$[Fe^{3+}]_{app.}$ (mM)
time (d)
F1
F2
F3
F4
F5
c
plate-to-flask
$AUC_{Fe^{3+}_{app.,AF}}$ repeatability
batch AUC deviation (%)
plates 0
flasks 0
flasks 1
flasks 100
$[Ga^{3+}]$ (mM)
d
plate-to-flask
$t_{max,Fe^{3+}_{app.,AF}}$ repeatability
batch timing deviation (%)
plates 0
flasks 0
flasks 1
flasks 100
$[Ga^{3+}]$ (mM)


a
OD600, app., (corr.) in plates = f(cells, t)
cell condition
Ab AF
A600
A*600
OD600 (AU)
time (d)
b
OD600, app., AF in flasks = f(t)
F1
F2
F3
F4
F5
planktonic OD600, app.
time (d)
c
plate and flask AUCOD600, app., AF
plate A600
plate A*600
flask OD600, app.
integrated response (% of control)
[Ga3+] (M)
d
plate and flask tmax, OD600, app., AF
plate A600
plate A*600
flask OD600, app.
delay to maximum (d)
[Ga3+] (M)


a
$Fe^{3+}_{app.,AF}$ in flasks
$[Fe^{3+}]_{app.}$ (mM)
time (d)
[Ga^{3+}] (mM)
0 1 ·0 ·1 ·6 ·12 ·18
[citric acid] (mM)
b
$Fe^{3+}_{app.,AF}$ in plates
$\Delta A_{380,net}$ (AU)
time (d)
c
$Fe^{3+}_{app.,Ab}$ in plates
$\Delta A_{380,Ab}$ (AU)
time (d)
d
$Fe^{3+}_{app.,\Delta AF}$ in plates = f(A_λ)
$\Delta A_{\lambda,net}$ (AU)
[citric acid] (mM)
[Ga^{3+}] (mM)
0 1 λ (nm)
380
428
e
plate-to-flask $AUC_{Fe^{3+}_{app.,AF}}$
$\log_2$ ($AUC_{rel,p}$/$AUC_{rel,f}$)
[plate additive] (M)
Al^{3+}
Cu^{2+}
Ga^{3+}
NaCl
VO^{2+}
citric acid, 0 mM Ga
citric acid, 1 mM Ga
equal dose
nearby doses
f
plate-to-flask $t_{max,Fe^{3+}_{app.,AF}}$
$\log_2$ ($t_{max,p}$/$t_{max,f}$)
[plate additive] (M)
final-read maximum
g
$Fe^{3+}_{app.}$ in plates = f(cells, [additive], t)
ΔA_{380} (AU)
time (d)
AF at LF endpoint
Ab LF medium
HH
+ ~10^{-2} M In^{3+}
+ ~10^{-1} M Ga^{3+}
h
$Fe^{3+}_{app.}$ in tubes = f(cells, [additive], t)
ORP (mV)
time (d)
Ab LF medium
HH
+ 10^{-2} M In^{3+}
+ 10^{-1} M Ga^{3+}
i
$OD_{600,app.,endpoint}$ in tubes = f(cells, [additive])
planktonic $OD_{600,app.,endpoint}$
Ab
LF
HH (baseline)
+ 10^{-2} M In^{3+}
+ 10^{-1} M Ga^{3+}


a
pH app., S2O3 2− in plates = f(cells, OD600, t)
Ab AF OD600
10−3
10−2
10−1
Rt − R0
time (d)
b
pH app., S4O6 2− in plates = f(cells, t)
Ab
AF
Rt − R0
time (d)
c
pH app. in plates = f(cells, energy source, t)
Ab AT 20 mM
S4O6 2−
S2O3 2−
Rt − R0
time (d)
d
pH app., corr., S2O3 2− in plates = f(cells, OD600, t)
Ab AF OD600
10−3
10−2
10−1
R*t − R*0
time (d)
e
pH app., corr., S4O6 2− in plates = f(cells, t)
Ab
AF
R*t − R*0
time (d)
f
pH app., corr. in plates = f(cells, energy source, t)
Ab AT 20 mM
S4O6 2−
S2O3 2−
R*t − R*0
time (d)
g
pH app., corr., S4O6 2− in tubes = f(cells, t)
Ab
AF
AT
pH
time (d)
h
OD600, app., endpoint in tubes = f(cells)
planktonic OD600, app., endpoint
Ab
AF
AT